

1. a. Using MongoDB, create a collection called transactions in database usermanaged (drop if it already exists) and bulk load the data from a json file, transactions. Json.
- b. Upsert the record from the new file called transactions_upsert. json in Mongoddb.

Execution Steps 1a:

1. In the terminal run, mongod - -version and mongosh - -version
2. Set the environment variables for the system to recognize the commands
3. Now run, mongosh command and enter
test> use usermanaged
usermanaged> db.createCollection("transaction") and exit
4. After creating Database and Collection from the terminal window, create a JSON file called transaction.json by clicking on File->New File in the current directory.
5. After the creation of transaction.json document file, import the document into the terminal window.
6. Now change the path from current directory to the mongodb database tool folder to recognize the mongoimport command using cd command
7. To import run, ./mongoimport - - db usermanaged - - collection transaction - - file "<path to transaction.json file>" - - jsonArray
8. Goto new terminal window and run, mongosh command
test> use usermanaged
usermanaged> db.transaction.find().pretty()
9. Now we can observe that the bulk data from json file is being displayed onto the terminal window.

transaction.json

```
[  
  {  
    "_id":1,  
    "name":"Anil",  
    "dept":"AIML",
```

```
"marks":80
},
{
  "_id":2,
  "name":"Sagar",
  "dept":"CSE",
  "marks":90
}
]
```

Execution Steps 1b:

1. Create a new JSON file called transactions_upsert.json by clicking on File->New File in the current directory.
2. After the creation of transaction_upsert.json document file, import the document into the terminal window.
3. In the terminal window, change the path from current directory to the mongodb database tool folder to recognize the mongoimport command using cd command.
4. Now run, ./mongoimport --db usermanaged --collection transaction --file "< path to transactions_upsert. json file >" --jsonArray --mode upsert
5. Goto new terminal window and run, mongosh command and enter
test>use usermanaged
usermanaged>db.transaction.find().pretty()
6. Now we can observe that the new record from the transaction_upsert.json file is being displayed onto the terminal window.

Note: Write down the output of the program 1a and 1b in the record on the left side.

2. Query MongoDB with Conditions: [Create appropriate collection with necessary documents to answer the query]

- a. Find any record where Name is Somu
- b. Find any record where total payment amount (Payment. Total) is 600.
- c. Find any record where price (Transaction. Price) is between 300 to 500.

Calculate the total transaction amount by adding up Payment. Total in all records.

Program:

// 1. Setup sample data

```
db.sales.insertMany([  
  
  { Name: "Somu", Payment: { Total: 600 }, Transaction: { Price: 400 } },  
  
  { Name: "Alice", Payment: { Total: 800 }, Transaction: { Price: 350 } },  
  
  { Name: "Bob", Payment: { Total: 200 }, Transaction: { Price: 200 } }  
  
]);
```

// 2. Find any record where Name is Somu

```
db.sales.find({ Name: "Somu" });
```

// 3. Find any record where total payment amount is 600

```
db.sales.find({ "Payment.Total": 600 });
```

// 4. Find any record where price is between 300 to 500

```
db.sales.find({ "Transaction.Price": { $gte: 300, $lte: 500 } });
```

// 5. Calculate the total transaction amount

```
db.sales.aggregate([ { $group: { _id: null, TotalAmount: { $sum: "$Payment.Total" } } } ]);
```

Execution Steps:

1. In the terminal run, mongod - -version
2. Then check for, mongosh --version
3. Now run, mongosh

```
test> use usermanaged
```

```
usermanaged>db.createCollection("sales")
```

4. Type the above given program queries inside the sales collection.
5. Now observe the output for each of the queries.

Note: Write down the output of the program in the record on the left side.